Tectonophysics, 68 (1980) 339—348
Elsevier Scientific Publishing Company, Amsterdam — Printed in The Netherlands

ANOMALOUS PALEOMAGNETIC DIRECTIONS IN LATE TERTIARY ANDESITIC INTRUSIONS OF THE CAUCA DEPRESSION, COLOMBIAN ANDES

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(Received June 4, 1979; revised version accepted January 17, 1980)

ABSTRACT

MacDonald, W.D., 1980. Anomalous paleomagnetic directions in Late Tertiary andesitic intrusions of the Cauca depression, Colombian Andes. Tectonophysics, 68: 339—348.

Late Tertiary hypabyssal intrusions of the northern Cauca depression in the northwest Andes of Colombia provide anomalous paleomagnetic declinations of approximately northwest trend. Individual tectonic rotation mechanisms of folding or faulting alone are incapable of accounting for the observed dispersed inclinations in the northwestern trend. Combinations of mechanisms, including regional shear, folding, and thrusting, may explain the anomalous paleomagnetic vector directions.

INTRODUCTION

Grosse (1926) made a detailed geologic study of the northern Cauca or Antioquia Basin, southwest of Medellin, in a coal resources study for the Antioquia Railroad. That classic work remains one of the best published references on the geology of the Tertiary Cauca basin of northwest Colombia. The present study summarizes results of a paleomagnetic survey of young porphyritic intrusions mainly from the area mapped by Grosse.

GEOLOGIC AND TECTONIC SETTING

The Cauca basin or "depression" is an unusual graben-like structure separating the Cordillera Occidental from the Cordillera Central of the Colombian Andes (Fig. 1). This structure is unusual in that it is bounded both east and west by high-angle faults thought to involve both thrusting and large lateral movements. This compressed basin is bounded on the east by mainly older crystalline rocks, intruded by Mesozoic batholiths. On the west,

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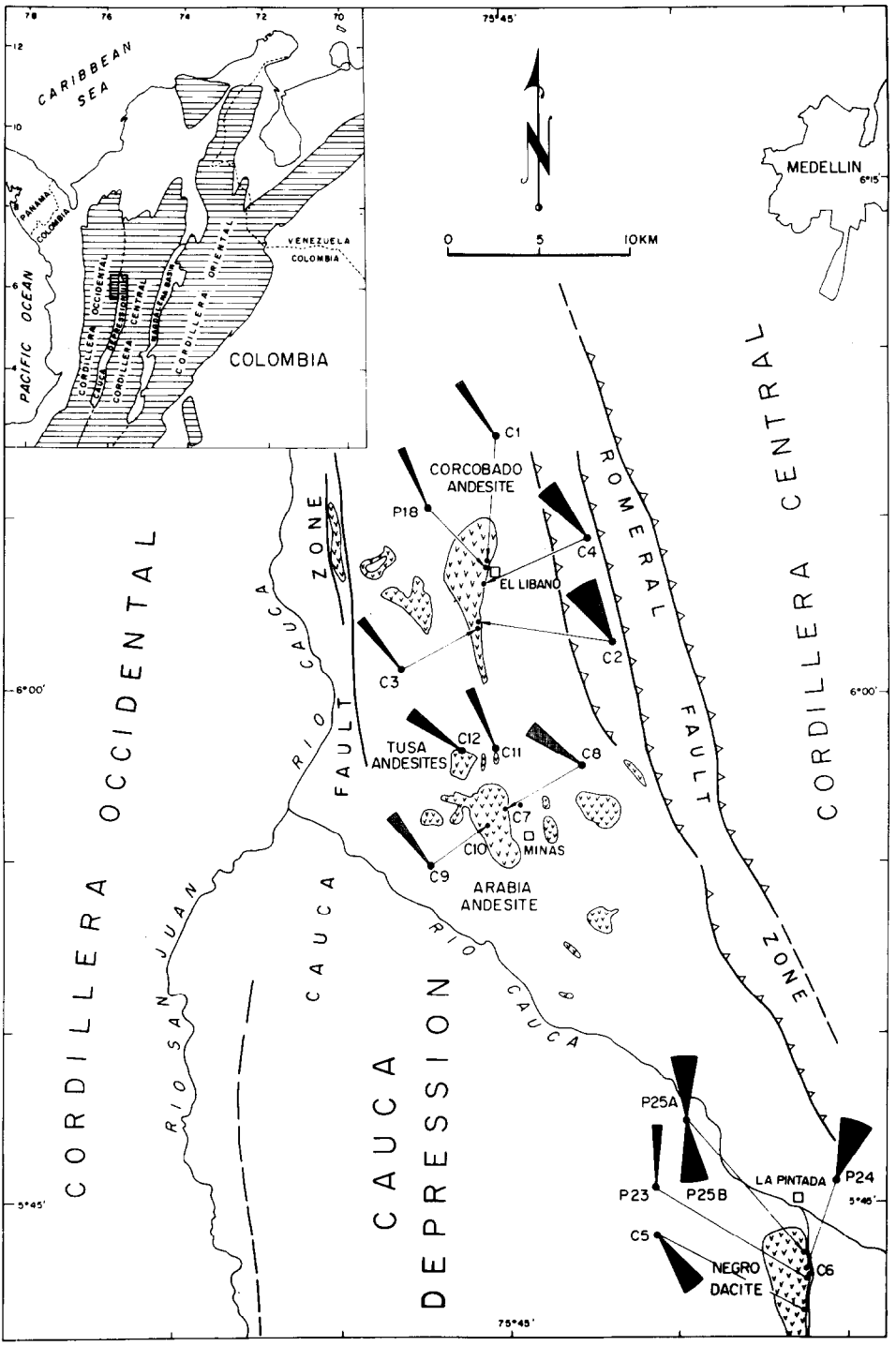


Fig. 1. Site location map, northern Cauca basin, Colombia. Wedges show paleomagnetic declination uncertainties about site mean directions in Late Tertiary porphyritic intrusions. Wedges represent horizontal projections of cones of 95% confidence. Anomalous northwest declinations are predominant. Index map in upper left corner shows location of survey area in Colombian Andes. After Grosse (1926).

it is bounded by deformed mainly Mesozoic rocks of probable oceanic origin. High-angle faults to the east, the Romeral fault zone, and to the west, the Cauca fault zone, accompanied the deformation of this basin in Late Miocene and younger time.

In his detailed study, Grosse (1926) concluded that high-angle reverse thrusts of steep easterly dip characterize the Romeral fault zone. Subsequent tectonic analyses (Barrero et al., 1969; Case et al., 1971; Campbell, 1974) have extended the term Romeral to include a major system of faults along the east margin of the Cauca basin. The prevailing sense of motion on this fault system seems to be right-lateral. On the Cauca fault zone, Grosse (1926) also inferred reverse thrust movements, of offset unknown.

The Romeral fault system lies along the approximate trend of Late Tertiary volcanism in this area of the Andes (Barrero et al., 1969). It also marks approximately an earlier continental margin, of probable Permian age, against which younger crustal rocks have been accreted (Case et al., 1971; MacDonald, 1972).

The Late Tertiary volcanic deposits of the Romeral trend accumulated on a continental margin of Mesozoic age, rather than in an oceanic island-arc setting. This is clear from the scarcity of marine Tertiary deposits and the abundance of continental deposits in the Cauca depression.

The area of study lies in the diffuse tectonic boundary between two major plates — the South America and Nazca plates, and is close to the ill-defined triple junction of the Caribbean—South America—Nacza plates (see Van Andel et al., 1971; Rea and Malfait, 1974). An excellent summary of the regional geology of the northern Andes is given by Irving (1976).

Two units dominate the Tertiary section of the northern Cauca area: the Antioquia Formation and the Combia Formation (Grosse, 1926; Van der Hammen, 1960). These units unconformably overlie deformed Mesozoic and older units and are intruded by porphyritic andesite intrusions of Late Miocene age which form the focus of this study. The Antioquia Formation, of probable Oligo-Miocene age, contains mainly terrestrial clastic rocks, conglomerate, sandstone, and shale, with interbeds of coal. Unconformably above lies the Combia Formation, predominantly volcani-clastic (Grosse, 1926; Jaramillo, 1976). Tuff and related pyroclastic rocks are most abundant. Flow rocks, about one-third of the Combia section, are mainly basalt with lesser amounts of andesite. The unconformity between the Antioquia and Combia may correspond to a notable regional hiatus of mid-Miocene age recognized throughout northwest Colombia (Duque-Caro, 1972).

The hypabyssal intrusives of the Cauca basin are mainly hornblende andesites. Five bodies were sampled, of which four are located in the area of Fig. 1: the Corcovado, Tusa, Arabia, and Negro intrusives. A fifth body, the Marmato intrusion, was sampled at a single site 1 km east of Marmato, a town 25 km south of the area of Fig. 1.

Replicate K-Ar whole-rock age determinations on the Corcovado body give a Late Miocene age, 8 Ma (Table I). This age falls near the older end of a

TABLE I

Replicate whole-rock K-Ar age determination on Corcovado andesite, northern Cauca depression, Colombia (site P18, 06.1N, 075.8W).

scc Ar_{rad}/gm	Ar_{rad}/Ar_{tot}	%K	Age(Ma)
$0.596 \cdot 10^{-6}$	0.09	1.87	8.0 ± 0.9
$0.578 \cdot 10^{-6}$	0.08	1.86	7.8 ± 1.0

Constants:
$\lambda_\beta = 4.72 \cdot 10^{-1}\ y^{-1}$
$\lambda_e = 0.585 \cdot 10^{-1}\ y^{-1}$
$K^{40} = 1.19 \cdot 10^{-4}$ atom percent natural K
Analyst: Westwood Laboratories, N.J., 1971

span of age-dates, 1—9 Ma, reported for volcanic clasts in the upper Magdalena basin on the opposite, i.e. east, side of the Cordillera Central (Van Houten, 1976). It is thought that the rise of the Cordillera Central post-dates the emplacement of the Cauca valley intrusions. The presently active volcanic trend follows the axis of the Cordillera Central, evidently indicating an eastward shift of the locus of volcanism between approximately 8 Ma and the present, from the Cauca basin to the mid-line of the Cordillera Central. The shift of the volcanic axis coincided approximately with the uplift of the Cordillera Central.

PALEOMAGNETIC RESULTS

Oriented cores and a few block samples were taken at eighteen sites in five hypabyssal intrusions of the Cauca basin. Field orientations were by magnetic compass. The predominant rock-type is porphyritic hornblende andesite. The Tusa, Arabia, and Negro bodies are possibly volcanic necks, with Arabia and Negro representing multiple intrusions; Corcovado is thought to be a laccolith (Grosse, 1926; Anonymous, 1968; Jaramillo, 1976). The Negro intrusions are dacite (Jaramillo, 1976).

Laboratory specimens, measuring 2.5 cm diameter and 2.3 cm length, were measured on a slow-speed spinner fluxgate-gradiometer (Foster, 1966) with output to a minicomputer. All specimens were demagnetized at 50, 100, 150, 200, and 300 Oe peak fields with demagnetization apparatus of the type described by Roy et al. (1973).

At 15 of the 18 sites, a stable characteristic paleomagnetic direction was recovered (Table II). No structural tilt corrections were applied to the directions isolated, which therefore represent in situ demagnetized directions. Median destructive peak fields range from 50 to 300 Oe, being between 75 and 175 Oe for most specimens. NRM intensities vary from 10^{-3} to 10^{-6} emu/cm^3 with the average near 10^{-4} emu/cm^3. Most sites are normally mag-

TABLE II

Site and rock-unit mean directions of magnetization for Cauca intrusions

Site	Peak field	N	Sense	Mean direction				Paleomagnetic pole		
				incl.	decl.	α_{95}	k	lat. (N)	long. (W)	A_{95}
Corcovado Andesite (06.1N, 075.8W)										
C1	300	5	n	−14.9	325.0	3.1	610	52.5	173.3	
C2	100	3	n	+08.4	330.5	14.3	66	60.6	168.0	
C3	100	3	n	+10.8	320.9	3.3	1409	51.1	164.7	
C4	300	2	n	−07.6	320.1	9.9	644	48.9	178.8	
P18	150	13	n	+04.9	336.4	3.5	142	66.2	173.6	
Mean		5	n	+00.3	326.6	12	39	56.1	174.7	8
Tusa Andesite (06.0N, 078.5W)										
C11	100	5	n	−22.8	305.2	5.5	192			
C12	200	8	n	−23.0	335.9	3.7	222			
Mean		2	n	−23.7	320.5	—	17			
Arabia Andesite (05.9N, 075.7W)										
C8	100	5	n	+53.9	304.7	4.3	312			
C9	200	5	n	+60.0	322.9	2.3	1125			
Mean		2	n	+57.3	313.0	25	98			
Negro Dacite (05.7N, 075.6W)										
C5	100	4	r	−06.2	143.8	9.7	91			
P23	100	3	n	+26.3	005.2	7.0	311			
P24	200	3	n	+21.3	019.3	14.8	70			
P25A	100	2	n	+24.4	358.8	10.5	563			
P25B	100	3	r	+10.7	170.7	12.1	106			
Mean		5	m	+14.3	355.2	25	10			

N is number of samples for mean, n is normal, r is reversed, and m is mixed polarity sense. Incl. is inclination, decl. is declination, k is precision parameter. α_{95} is half-angle of cone of 95% confidence about site-mean direction. A_{95} is radius of circle of 95% confidence about unit-mean pole calculated from virtual geomagnetic poles.

netized; only the Negro body shows mixed polarities.

From Table II and Fig. 3, the paleomagnetic declinations for most bodies can be seen to be of northwesterly trend, the average being about 320 (N40W). Inclinations vary from +55° to −25°. These directions are anomalous for rocks of this age and location. The directions of the paleomagnetic vectors are believed to have been re-oriented tectonically, as discussed below.

Declination uncertainties, $\pm\beta$, about the mean declination have been plotted in Fig. 1. The declination uncertainty is obtained from (Fig. 2):

$$\beta = \arcsin(\sin \alpha_{95}/\cos I) \quad (1)$$

The directions isolated have mainly low inclinations and small α_{95}, and consequently small declination uncertainties. Declinations for the Corcovado,

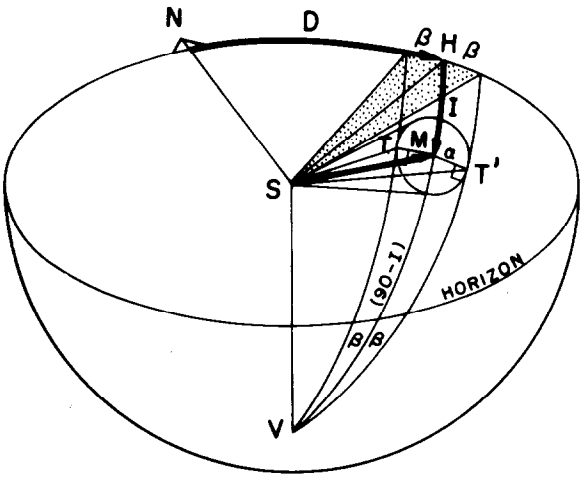


Fig. 2. Declination uncertainty, $\pm\beta$, about the mean declination, is shown as shaded wedge. Planes STV and $ST'V$ are vertical planes through site S tangential along ST and ST' to the cone of 95% confidence about the site-mean direction SM.

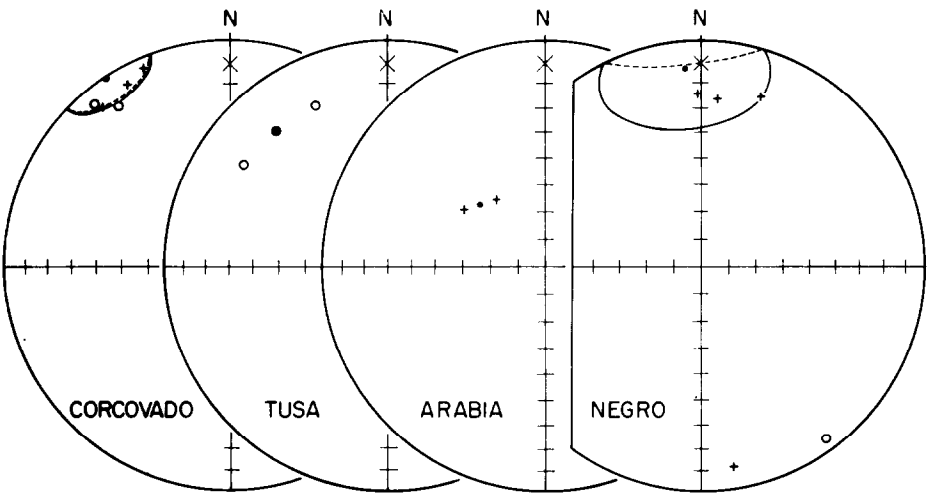


Fig. 3. Mean paleomagnetic directions for Late Tertiary intrusions of the northern Cauca depression are dispersed in vertical plane of NW trend. Small filled circles represent rock-unit means for the Corcovado, Tusa, Arabia, and Negro bodies, with circles of 95% confidence. Site means are indicated by plus-symbols (lower hemisphere) and by open circles (upper hemisphere). Direction of present axial geocentric theoretical dipole field is indicated by × symbol. Equal area projection, local horizontal as prime.

Tusa, and Arabia intrusions clearly trend northwesterly, about 35—40NW. The single Marmato site, not plotted in Fig. 1, has a similar declination, with an inclination near zero.

A paleomagnetic pole was calculated from site geomagnetic poles for the Corcovado body (Table II). At 8 Ma, this is the only body of relatively well-defined age. The corresponding pole is at 56N 185E, $A_{95} = 8°$, $N = 5$. The paleomagnetic poles for coeval rock units on the South American plate are not significantly different from the present geographic pole (Richardson and Watkins, 1967; Mazoires et al., 1971). Thus, tectonically undisturbed units for the Cauca valley should have paleomagnetic directions not perceptibly different from the present theoretical dipole field direction (+12°, 000°). The northwesterly declinations of the Cauca intrusions are thought to indicate large tectonic rotations late in the Andean orogeny. This aspect is explored further below.

TECTONIC IMPLICATIONS AND DISCUSSION

Paleomagnetic data for the Late Tertiary of South America are sparse. The Late Tertiary pole is thought to lie close to the present geographic pole. The Cauca directions are clearly not typical Late Miocene directions for South America, being unusual in declination, inclination, or both. The site mean vectors for the Corcovado, Tusa, and Arabia bodies lie approximately in a vertical plane of northwesterly trend. This unusual pattern can be only partly explained by a rotation of these bodies through 35° clockwise about a vertical axis. Assuming an age of 8 Ma, this indicates a rotation rate of 4°/Ma or faster. This is a relatively large rotation for rocks so young, and the counterclockwise sense is puzzling in view of the right-lateral sense of movement thought to characterize the Romeral zone. The dispersal of the paleomagnetic vectors in the vertical plane, from —25° to +55°, is yet more unusual. No acceptable single rotation mechanism could be found to decrease the dispersion of these vectors. The distribution suggests that the tectonic rotation involved other than vertical axes.

The rotation rates are rapid for lithospheric plate movements but favor domains of small size. Differences between rock-unit mean directions give an indication of the size of rotated domains. For example, the Corcovado, Tusa and Arabia directions are in agreement within themselves, but are different from each other. Spanning a distance of 10 km, these sites help to define domains of rotation of dimension smaller than 5 km in plan. A wide spectrum of domain sizes is thought to be present, large massive igneous bodies representing the larger domains, and disrupted bedded units representing the smaller domains.

The clustering of paleomagnetic vectors decreases from the between-specimen to the between-rock-unit level. Relative movements seem to be larger between rock units than within rock units. This conclusion is reasonable from a structural point of view. The massive igneous intrusions are expected

to behave mechanically as more competent than the intervening and surrounding stratified sections of well-bedded pyroclastic and sedimentary rocks. Thus it is predictable that, under tectonic stress, the Cauca intrusions should undergo movements relative to each other.

Although a normal paleomagnetic direction is obtained for the Negro body, the latter does not mark the southern limit of the rotated terrains. A site at Marmato, 25 km south of the area of Fig. 1, has also a northwest paleomagnetic declination, suggesting that the affected terrain extends at least 70 km along strike.

Tectonic rotations are to be expected in the northwest Andes of Colombia since many young faults, both major and minor, are known, including several major strike-slip faults (Feininger, 1970; Case et al., 1971; Van Andel et al., 1971; Campbell, 1974; Irving, 1976). The prevailing opinion of displacement on the strike-slip faults is right-lateral. If the sense of rotation as determined paleomagnetically can be related to regional shear (Irving, 1964; Beck, 1976), the prevailing sense of right-lateral shear in the Cauca area would predict clockwise rotations. The observed declinations suggest a counterclockwise rotation sense.

The Cauca directions seem to be unusual also in their inclinations as noted above. In tectonically rotated terrains of the Western Cordillera of North America, Beck (1976) has noted the tendency for inclinations to be more shallow than expected. In the Cauca terrain, inclinations are both more steep and more shallow than expected; a more detailed study would be required to clarify the trends here.

Various unsatisfactory attempts to find common factors in the rotation angles and axes were made. Since it is clear that the observed rock-unit directions cannot be explained solely by rotation about vertical axes, an attempt was made to explain the observed directions by rotations about axes normal to the Romeral faults. It was found not possible to rotate the theoretical dipole field direction into agreement with the observed directions by rotations about axes normal to the steeply east-dipping Romeral faults. A causative mechanism by rotation on faults is possible by movements on faults of various orientations. In particular, the observed directions can be explained by large rotations on faults of ENE strike and steep SE dip, an uncommon trend in this zone.

Assuming that the observed rock-unit mean directions result from rotations through varying angles about a common axis, that axis can be found. By drawing great circles representing all possible positions of rotation axes which could rotate the geocentric dipole field direction into the observed directions, one locates a common intersection of such great circles. In the Cauca case, the great circles intersect in a set of axis with shallow plunge and NNW trend, in agreement with the known trend of folding. While this suggests that the observed directions result from folding, this solution is regarded as improbable since the required rotations are much greater than bedding tilts observed in the north Cauca basin.

A third possibility, considered as more plausible, is that combinations of folding and thrusting have resulted in the apparent rotations. Many such tectonic combinations are possible, however, and common factors of rotation could not be isolated with the data available. In particular, it is desirable to know the in situ paleomagnetic directions in units displaying bedding.

Finally, anomalous paleomagnetic directions are known from other areas of the Colombian orogenic zones, such as the Guajira Peninsula in the Caribbean (MacDonald and Opdyke, 1972), the Cordillera Central southeast of the study area at Ibague (Scott, 1978), and possibly in the Cordillera Oriental (Creer, 1970). The present study of the north Cauca depression reveals that large tectonic re-orientations were caused by young Andean movements, which may still be underway. An expanded survey of the Cauca region would be worthwhile in understanding the tectonic patterns of this complex zone.

CONCLUSIONS

A glimpse of the tectonic complexity of Andean movements is provided by a paleomagnetic survey. Large and rapid tectonic rotations have affected young hypabyssal andesitic intrusions of the north Cauca depression in the northwest Colombian Andes. Paleomagnetic directions have been dispersed into a vertical plane of approximately N35W trend, suggesting that a combination of rotations, or more than one mechanism, is involved. The affected terrain extends at least 70 km along strike. A better understanding of the causative mechanisms will require more extensive sampling supported by geologic mapping in the Cauca depression and adjacent Andes.

ACKNOWLEDGEMENTS

This research was supported by grant DES75-15371/EAR77-09091 from the Division of Earth Sciences of the National Science Foundation, Washington, D.C. Paul Davis provided capable and resourceful assistance in both field and laboratory. E. Longo, D. Pfann, and E. Liebes assisted ably with the construction of equipment. Professor G. Botero Arango of the Universidad Nacional at Medellin provided a helpful introduction to the geology of the Medellin area.

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